

Long-term operating stability in perovskite photovoltaics

Hongwei Zhu^{1,6}, Sam Teale^{2,6}, Muhammad Naufal Lintangpradipto¹, Suhas Mahesh², Bin Chen², Michael D. McGehee³, Edward H. Sargent^{2,4,5}  & Osman M. Bakr¹ 

Abstract

Perovskite solar cells have demonstrated the efficiencies needed for technoeconomic competitiveness. With respect to the demanding stability requirements of photovoltaics, many techniques have been used to increase the stability of perovskite solar cells, and tremendous improvements have been made over the course of a decade of research. Nevertheless, the still-limited stability of perovskite solar cells remains to be fully understood and addressed. In this Review, we summarize progress in single-junction, lead-based perovskite photovoltaic stability and discuss the origins of chemical lability and how this affects stability under a range of relevant stressors. We highlight categories of prominent stability-enhancing strategies, including compositional tuning, barrier layers and the fabrication of stable transport layers. In the conclusion of this Review, we discuss the challenges that remain, and we offer a perspective on how the field can continue to advance to 25-year and 30-year stable perovskite solar modules.

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¹KAUST Catalysis Center, Division of Physical Sciences and Engineering, King Abdullah University of Science and Technology, Thuwal, Kingdom of Saudi Arabia. ²Department of Electrical and Computer Engineering, University of Toronto, Toronto, Ontario, Canada. ³Materials Science and Engineering Program, University of Colorado Boulder, Boulder, CO, USA. ⁴Department of Chemistry, Northwestern University, Evanston, IL, USA. ⁵Department of Electrical and Computer Engineering, Northwestern University, Evanston, IL, USA. ⁶These authors contributed equally: Hongwei Zhu, Sam Teale. ✉ e-mail: ted.sargent@northwestern.edu; osman.bakr@kaust.edu.sa

Introduction

Because of their remarkable properties, metal-halide perovskite (MHP) semiconductors have ushered in a new era of solar cell research. MHPs form an ABX_3 structure in which the A site is a monovalent cation (typically a mixture of caesium, methylammonium or formamidinium), the B site is a divalent metal (usually Pb or Sn) and the X site is a pure halide or mixture of halides (I, Br, Cl) (Fig. 1a). Owing to their large absorption coefficients, high mobilities, long diffusion lengths and tunability, polycrystalline thin films of perovskite have been used to fabricate perovskite solar cells (PSCs) with bandgaps from 1.2 to 2.3 eV (refs. 1,2) and power conversion efficiencies (PCEs) approaching 26% (~ 1.5 eV)³. Metal-halide perovskites are commonly known to have intrinsic defect tolerance, but modest temperatures (≥ 60 °C) can cause perovskite crystal decomposition^{4,5}, and 1-sun illumination can break bonds to induce degradation^{6,7}. Unsurprisingly, the time taken for early PSCs to drop to 80% of their original performance (their T_{80} lifetime) was only a few hours⁸. Since then, great strides have been made, and some PSCs have demonstrated thousands of hours of operational stability at ≥ 65 °C with negligible PCE loss^{9,10}. Despite these improvements, the fundamental understanding of instabilities and the mechanisms behind stability-enhancing methods are often overlooked.

In this Review, we focus on lead-based perovskites used in single-junction PSCs. First, we discuss the special properties of perovskite materials and the major advances in PSC stability. Then we discuss the factors affecting PSC stability, the techniques used thus far to combat them, and potential routes to further improvement. Several reports have demonstrated encapsulation that renders humidity and oxygen effects negligible on timescales up to 1 year (ref. 11). Hence, it seems reasonable to consider moisture and oxygen ingress separately as an encapsulation issue. The need for robust, long-term encapsulation techniques and the best ways to encapsulate PSCs are discussed at length elsewhere¹². Here, we instead focus on illumination, bias and temperature-based degradation.

Differences between perovskites and traditional semiconductors

Perovskites and traditional semiconductors (such as Si, CdTe and III–V semiconductors) have several key differences (Fig. 1a,b) that ultimately affect their stability. Paramount to the success of perovskites is their defect tolerance, a result of their band structure (Fig. 1a, right). In typical semiconductors, the valence band maximum (VBM) is formed from bonding orbitals and the conduction band minimum (CBM) from antibonding orbitals¹³. When these bonds are broken, dangling bonds form close to where the original bonds were, creating defects deep within the semiconductor bandgap (Fig. 1b, right). Perovskites instead form their bandgap from antibonding orbitals at both the VBM and CBM, and thus breaking these bonds produces states away from the bandgap, either shallow defects or states within the valence band. The upshot of this is that polycrystalline perovskite films with defect densities 10^6 times greater than single-crystal silicon can produce similar solar cell performance¹⁴. The downside is that high defect densities reduce the energy needed for ions to migrate through the film, and defects tend to propagate over time, resulting in faster degradation of the absorber¹⁵.

A second important difference between perovskites and traditional semiconductors is the lone pair effect. The divalent metal B site (usually Pb or Sn) forms with a lone pair of electrons that dictates the local geometry of the perovskite lattice¹⁶ (Fig. 1c). Similarly to how a water molecule is bent by the lone electron pairs on the oxygen, the perovskite lattice is distorted by the lone pair on the B site. However, rather

than distorting to one specific orientation (like in the case of water), several local minima exist, and at room temperature the perovskite lattice is continually moving and reorienting. This behaviour is not clearly resolved by X-ray diffraction, which reveals the average lattice spacing, but is obvious in the Raman spectra of perovskites, which at room temperature appear similar to fluids¹⁷ (Fig. 1d). The dynamic nature of the perovskite lattice has many profound consequences for its stability. The thermal expansion coefficients of metal-halide perovskites are typically 10 times that of silicon, which is an important consideration when evaluating the mismatch in expansion and thus strain between perovskite and contacts. Additionally, the dynamic lattice increases the density of interstitials and vacancies, and aids ion migration¹⁶.

Ion migration is of great consequence in perovskite photovoltaics. High defect densities generate high ion densities (up to 10^{17} cm⁻³) (ref. 18) and provide pathways for ion migration through the lattice or along grain boundaries. The ion migration energy of halides is lower than for cations or metal species¹⁹, meaning halide drift is generally attributed as the reason that perovskites act as both electronic and ionic conductors. PSCs are typically made by sandwiching a mostly intrinsic perovskite semiconductor layer between an electron-transport layer (ETL) and a hole-transport layer (HTL). Mobile ions in the perovskite migrate to the interfaces until they screen the built-in electric field that would normally exist when a semiconductor with a relatively low carrier density is between electrodes with different work functions. If the voltage across a PSC is changed so quickly that the ions cannot redistribute fast enough, then, depending on the direction of the voltage scan, the field can either be larger than normal (which favours electron extraction) or it can point in the wrong direction (which hinders electron extraction)¹⁸. On initial testing, hysteresis is measured between forward and reverse current–voltage (J – V) scans with strong dependency on scan speed^{20,21}. In the long term, ion redistribution results in a loss of current over time due to either flattening bands at the device interface or a build-up of insulating interface material²². More troublingly, we now know that in iodide-based MHPs, the density of mobile ions is increased under illumination. As the VBM is formed from iodide bonds, photogenerated holes could oxidize iodide to make neutral iodine interstitials and iodide vacancies (Fig. 1e) causing several knock-on effects: iodine interstitials act as recombination centres and, because iodide vacancies are conductive, they move towards the interfaces^{23,24}, I_2 gas is produced and can leave the perovskite, and iodine ions diffuse into charge-transport layers (CTLs) where they can react with metal electrodes, reducing conductivity²⁵ (Fig. 1f).

Another phenomenon that affects perovskite stability is phase segregation under illumination. For most perovskites in the family $AB[I_{1-x}Br_x]_3$, iodine and bromine can be mixed uniformly for x ranging from 0 to 1. However, typically when x is greater than 20–33%, the perovskite phase separates into iodide-rich and bromide-rich regions under illumination²⁶. The characteristic demonstration of this phase segregation is redshifted photoluminescence. When the film absorbs light, electrons and holes migrate to (lower-energy) iodide-rich regions, and as segregation progresses, the photoluminescence redshifts over time (Fig. 1g). The fundamental driver for segregation is debated, some suggesting it is thermodynamic in origin^{27,28}, others suggesting it is the result of polaron-induced lattice strain^{29,30} or charge carrier gradients^{31,32}. Some consensus is emerging that iodide is oxidized by photogenerated holes and has a higher mobility than bromine, and this sets up the opportunity for halide segregation^{33,34}. Regardless, phase segregation limits the potential open-circuit voltage (V_{oc}) of ≥ 1.7 -eV-bandgap PSCs necessary for tandem photovoltaics.

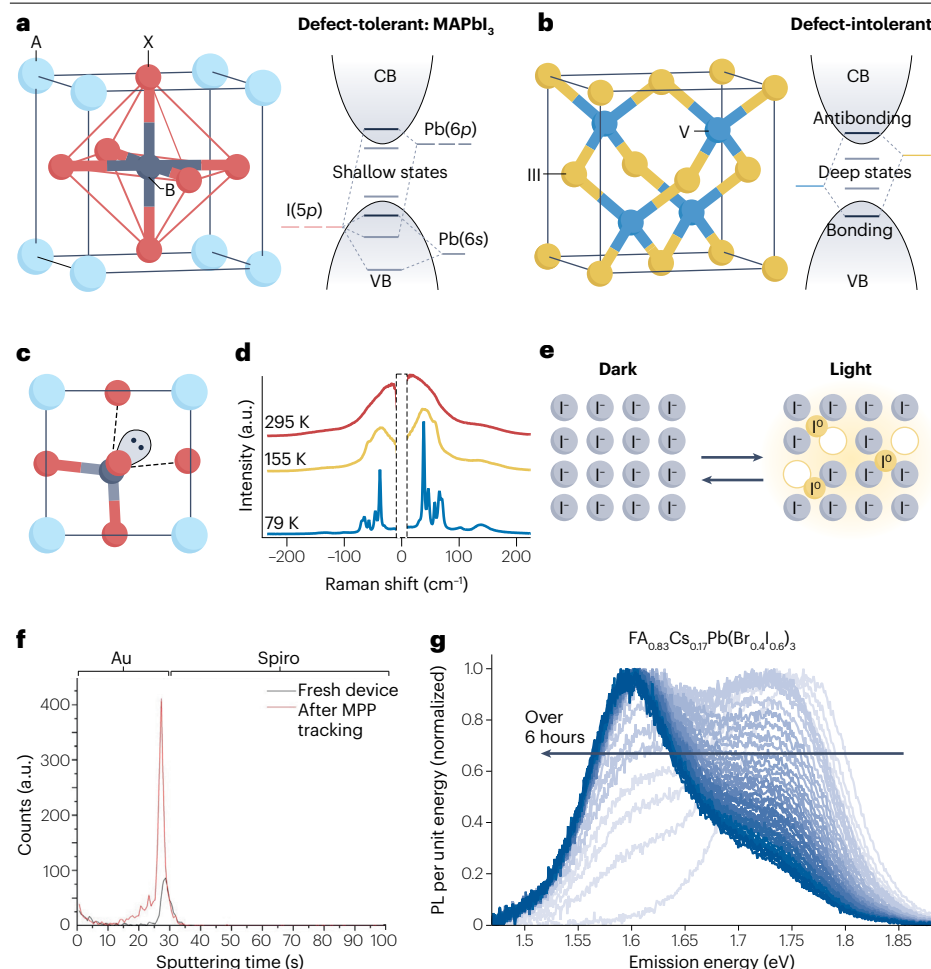


Fig. 1 | The unusual properties of perovskite semiconductors. **a**, The crystal structure of perovskite semiconductors (left) and the accompanying band structure (right). **b**, The crystal structure of traditional III–V semiconductors (left) and the accompanying band structure (right). **c**, Illustration of lattice distortion due to the lone pair effect in perovskite semiconductors. **d**, Temperature-dependent Raman spectra of MAPbBr₃ demonstrating the fluid-like bonding nature of metal-halide perovskites at room temperature. **e**, Schematic of light-induced ionic restructuring in iodine-based perovskites. **f**, Time-of-flight secondary ion mass spectroscopy showing a build-up of iodine ions at the metal/hole-transport layer interface after ageing a perovskite solar cell. The build-up is due to halide diffusion through charge-transport layers. **g**, Photoluminescence (PL) measurements of a 40% Br film, showing the increase of emission from iodine-rich regions as a function of illumination time. CB, conduction band; MPP, maximum power point; VB, valence band. Panel **a**, right, adapted from ref. 13, CC BY 4.0. Panel **b**, right, adapted from ref. 13, CC BY 4.0. Panel **c** adapted from ref. 16, Springer Nature Limited. Panel **d** adapted with permission from ref. 17, APS. Panel **e** adapted with permission from ref. 195, Max Planck Institute for Solid State Research. Panel **f** reprinted with permission from ref. 196. Copyright 2019 American Chemical Society. Panel **g** reprinted from ref. 197, CC BY 4.0.

Device stability so far

The PCE of PSCs now rivals that of Si photovoltaics (PV), and thus device stability is of utmost importance. The stability of PSCs depends on many factors (ageing condition, perovskite composition, CTLs, electrodes, encapsulation and passivation) and is thus a complex issue. Considering ageing conditions alone, a consensus statement on stability testing of PSCs was made in 2020 (ref. 35), and researchers are encouraged to use the ISOS testing protocols to enable comparison between studies. PSCs are classified into two architectural configurations: conventional (n–i–p) and inverted (p–i–n) structures (Fig. 2a,b). Figure 2c shows a timeline of PSCs, comparing record certified efficiency in n–i–p and p–i–n cells along with reported stability. We find that even though the efficiency of the record n–i–p devices has improved hugely in the past decade, their stability has improved relatively slowly (Box 1). Some reports on record devices are without stability measurements, others use different device fabrication with lower efficiency for stability testing and those that use the same architecture record T_{80} lifetimes at room temperature of only a few hundred hours³⁶. These results give the impression that poor stability is abundant among the best PSCs. Scrutinizing further, one finds that the situation is more positive. The stability of record p–i–n devices, for example, has vastly improved, with devices demonstrating stability under illumination and heat (ISOS-L2/3) with

$T_{80} > 1,000$ h from initial PCEs >23%. It should be noted that similarly high stability has been demonstrated in n–i–p, but not with record efficiency^{8,10,22}. Indeed, we also notice that many impressive stability milestones have been achieved. These are generally accomplished with low-efficiency devices first, and a few years is taken to achieve the same stability in a device with record efficiency.

Factors other than ageing condition that affect perovskite stability are often considered separately from efficiency, with many reports using CTLs or metal contacts for their stability testing that are different from those used to obtain their best PCEs. This is particularly popular in n–i–p devices where the hole-transporting material (HTM) spiro-OMeTAD is replaced with more stable HTMs^{37–40}. This causes two problems: stability is achieved with significantly lower (often unreported) PCEs than the best efficiencies, and cells are optimized for efficiency using spiro-OMeTAD when these optimizations may not be beneficial using other CTLs. As the field matures, it is important to combine stability with efficiency, considering how best to optimize devices within a stable architecture.

Another development is outdoor perovskite stability testing. Initially, real-world testing produced T_{80} lifetimes of only a few days⁴¹. More recently, however, the use of lamination-based glass–glass encapsulation has produced cells with no PCE degradation after

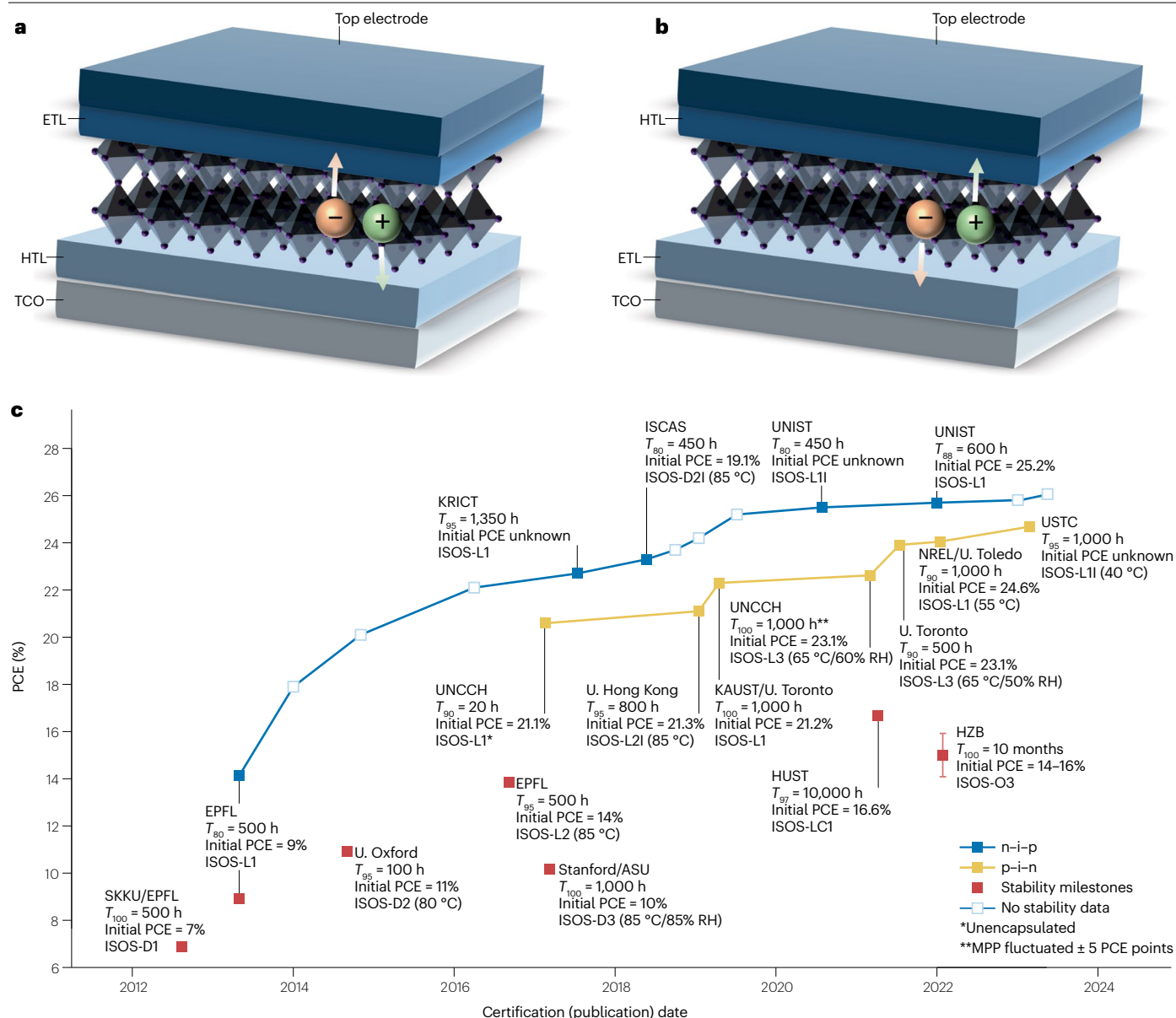


Fig. 2 | Timeline for stability in perovskite solar cells. a,b, The configuration of typical p-i-n (a) and n-i-p (b) perovskite solar cells (PSCs). **c,** A timeline for stability among PSCs with record efficiency. The blue and yellow lines represent record certified efficiency (steady-state) in n-i-p and p-i-n, respectively, whereas the red squares denote the earliest report of certain stability milestones: D1, dark storage (room temperature); D2, dark storage (high temperature); D2I, L1I and L2I, corresponding tests carried out in

an inert atmosphere (intrinsic stability testing); D3, dark storage (high temperature + high humidity); L2, maximum power point (MPP) tracked under illumination (high temperature); L2C, light-cycled (room temperature); O3, outdoor testing at MPP. ETL, electron-transport layer; HTL, hole-transport layer; PCE, power conversion efficiency; RH, relative humidity; TCO, transparent conductive oxide. References can be found in Supplementary Table 1.

10 months of maximum-power-point (MPP) tracking outdoors in Berlin⁴² (Fig. 2c). A small-scale (~3-m² active area) perovskite solar farm with lamination encapsulation and an initial efficiency ~12.5% PCE was tested over the course of a year, with a recorded T_{80} of 5,832 h (8 months) following the ISOS-O3 protocol⁴³. These results indicate that the lifetime of PSCs is increasing beyond what is practically assessable, and there is a need to move from standard test conditions towards accelerated testing.

Factors affecting PSC stability

To understand the rapid increase in PSC stability and how to improve it further, we must first understand which materials and stressors contribute. Despite differences in configuration between n-i-p and p-i-n devices, both consist of similar layers: the electron-transporting layer (ETL), hole-transporting layer (HTL), perovskite light-absorbing layer, cathode and anode contacts (Fig. 2a,b). Under different external stressors (that is, illumination, thermal and bias), the instability

of each constituent layer and the interfaces between layers are both important factors leading to operational instability.

Perovskite stability

Inherent stability of perovskite structure/phase. Before delving into the stability of full PSCs, we should examine the intrinsic stability of the perovskite structure. The majority of highly efficient PSCs are based on a 3D perovskite in a cubic crystal phase (also known as the α -phase). The Goldschmidt tolerance factor τ can be used to

determine the stability of a given perovskite from the ionic radii of its constituent ions:

$$\tau = (R_A + R_X) / \sqrt{2(R_B + R_X)},$$

where R_A , R_B and R_X are the ionic radii of the corresponding ions. In general, if τ for a given perovskite composition is between 0.9 and 1.0, it forms an ideal cubic structure. If τ is between 0.71 and 0.9, it will form an orthorhombic structure around room temperature. For MHPs,

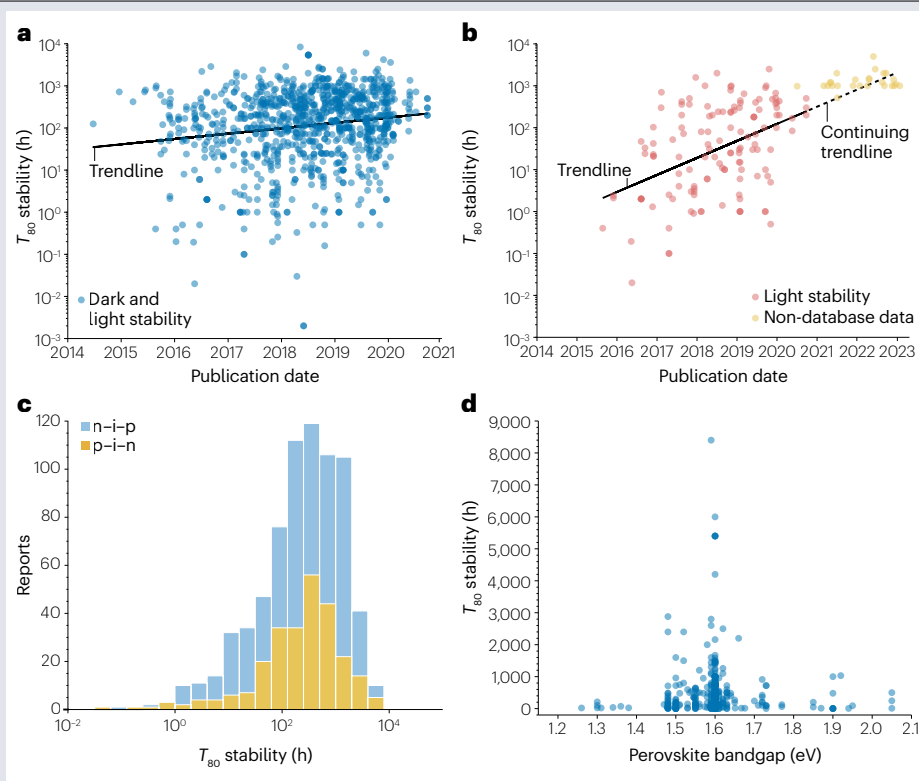
Box 1

The Perovskite Database

An open-access database of perovskite solar cell (PSC) results has been generated with data from >40,000 devices published between 2012 and 2020 (ref. 198), most of which have no stability data associated with them. Still, there are >1,000 devices with stability referenced to the ISOS protocols, which we used to generate the figure.

Panel **a** of the figure compares stability over time from all PSCs in the Perovskite Database with corresponding ISOS protocols. The data are understandably noisy, as there are various perovskite compositions, device architectures and ageing parameters included. The trendline still demonstrates an increase in T_{80} lifetime from ~10 h to ~100 h over 6 years. However, noise brought about by the huge volume of studies under-represents the stability gains. Looking instead at only PSCs aged under illumination (see the figure, panel **b**), the increase in stability from 2015 to 2020 becomes more obvious, probably because this is a more intense ageing test in which early devices fared poorly. To update the Perovskite Database statistics, we added 26 data points from recent high-impact studies from 2021 onwards (Supplementary Table 2). From these data, we can see that in modern studies, 1,000 h operational stability is readily achievable. This higher stability becomes particularly obvious when one realizes that only 4 of the 26 devices decayed to 80% of their original efficiency during stability testing. For the other 22, the number recorded is either the T_{90} , T_{95} or T_{100} lifetime. Indeed, with lifetimes extending well beyond 1,000 h, researchers are turning to accelerated ageing to assess stability, evidenced by the fact that 15 of the 26 devices were tested above room temperature (45 °C to 85 °C).

Panel **c** of the figure compares n-i-p and p-i-n cell stability. Considering cell configuration, we find that (when normalized to ISOS-L1 conditions using a simple acceleration factor model discussed in Supplementary Note 1) there is little difference in stability on average between n-i-p and p-i-n devices, which is surprising, as many state the stability of transport layers in p-i-n



as an advantage of the configuration^{141,158,199}. Still, this equivalence is bolstered when considering our 26 recent data points with high stability, which are evenly spread between n-i-p and p-i-n (Supplementary Fig. 1). Finally, we find that the vast majority of PSC stability is reported in cells with bandgaps from ~1.5 eV to 1.65 eV (panel **d** of the figure). This result is unsurprising as most PSC research is in this range, but demonstrates the need for more work in the areas of low-bandgap (~1.2 eV Pb/Sn) and wide-bandgap (~1.8 eV) cells that are used in all-perovskite tandems. Although not the subject of this Review, it should be noted that in Pb-free, Sn-containing perovskites, the oxidation of Sn(II) to Sn(IV) results in poorer stability for narrow-bandgap PSCs²⁰⁰. Additionally, higher ionic conductivity and halide segregation hamper the stability of wide-bandgap films with >20% Br concentration²⁰¹.

it has been observed that if $\tau > 1$ or $\tau < 0.8$, then the lattice distortion generated induces a photo-inactive, non-perovskite structure to form, normally known as the δ -phase⁴⁴.

MAPbI₃ has a τ value of -0.9 and an ideal cubic structure at room temperature. FAPbI₃ (τ of -1) is near the upper boundary for a cubic structure, so it has two phases at room temperature: cubic (α -phase) and hexagonal (δ -phase). High-temperature annealing (>433 K) is required to obtain α -phase FAPbI₃ (ref. 45), but δ -phase formation only needs a temperature of -285 K (ref. 46). Because the δ -phase is more stable than the α -phase, it is difficult to stabilize cubic FAPbI₃. By contrast, τ is approximately 0.8 for CsPbI₃, which is close to the lower range for the cubic perovskite structure. The thermodynamically stable phase of CsPbI₃ at room temperature is the non-photoactive δ -phase, and the photoactive α -phase is obtained by high-temperature annealing of the δ -phase (>633 K)⁴⁷.

Perovskite degradation under thermal stress. Some perovskite materials will undergo decomposition reactions under thermal stress. It is important to consider both operational temperatures (-40 °C to 75 °C; a temperature of 65 °C is commonly used as the upper end of PSC operating temperature, but there is evidence that this could be as high as 70–75 °C (refs. 43,48) and temperatures >75 °C for accelerated degradation testing. Note that temperatures used for accelerated ageing should not introduce a new decay mode that would not be observed below 75 °C. MAPbX₃ (X = Cl, Br, I) have low decomposition temperatures (<100 °C), with MAPbI₃ in particular degrading within the operational range (-60 °C), which is likely to preclude its use as a stable PSC material⁴⁹.

The thermal stability of FAPbI₃ is reportedly higher than MAPbI₃. FAPbI₃ films exposed to 150 °C for 60 mins are stable without discoloration, whereas MAPbI₃ discolours after -30 minutes⁵⁰. This enhanced stability is attributed to the stronger interaction between FA⁺ and [PbI₆]⁴⁺ compared with MA⁺ (ref. 51). Despite this, FAPbI₃ will undergo thermal decomposition at relatively low temperatures (>50 °C), where formamidinium iodide (FAI) decomposes into formamidine and HI⁵² (Fig. 3a). Both of these reactions are reversible and in a well-encapsulated device might not be fatal. When the temperature is around 95 °C, formamidinium decomposes into *sym*-triazine and ammonia through irreversible reactions⁴⁵ (Fig. 3a). When the temperature exceeds the phase transition temperature of α -FAPbI₃ (160 °C), FAPbI₃ partially decomposes into PbI₂ as FAI evaporates to HCN and NH₃ (ref. 53). This degradation behaviour suggests that FAPbI₃ has potential as a stable PSC material between -40 °C and 75 °C, although accelerated ageing above 95 °C is likely to be non-representative.

Because CsPbI₃ does not contain volatile and decomposable organic components, it does not break down at low temperatures, and α -CsPbI₃ is stable at temperatures as high as 390 °C. Above this, PbI₂ volatilizes and leaves CsI as residue⁵⁴. Hence, accelerated ageing beyond 300 °C is feasible, although the stability of other PSC layers would come into question.

Perovskite degradation under illumination. Ultraviolet and blue wavelengths have the strongest impact on PSC stability⁵⁵. Perovskites rapidly decompose under illumination when oxygen is present, although without oxygen, illumination can still trigger decomposition through simultaneous reversible and permanent reactions^{56,57}. Our discussion will focus on illumination without oxygen, assuming high-quality encapsulation.

PbI₂ films degrade into metallic Pb⁰ and I₂ gas under simulated sunlight (Xe lamp), with an activation energy (E_a) of -9 kcal mol⁻¹, or

0.4 eV per Pb atom⁵⁸. The activation energy is increased by a factor of 6 (-57 kcal mol⁻¹) on using a white-light LED. The degradation is retarded completely for photon energies less than 2.3 eV (ref. 59), which is the bandgap of PbI₂, suggesting that degradation is caused by light absorption. For MAPbI₃ under a Xe lamp, the E_a for Pb⁰/I₂ decomposition is lower than PbI₂ (-6 kcal mol⁻¹) owing to the lower bandgap of MAPbI₃ and [PbI₆] octahedral distortion which produces shorter I-I bond distances, aiding the release of I₂ (ref. 58). Moreover, when MAPbI₃ is illuminated with energies higher than 450 nm, signals from CH₃NH₂ and H₂ are detected, indicating that the N-H bond in the perovskite can dissociate under sunlight⁶⁰ (Fig. 3b). Although most studies examining degradation under illumination have been carried out on MAPbI₃, decomposition into Pb⁰ has been witnessed in FACs perovskites also⁶¹.

Most perovskites with a >20% Br to I ratio will segregate into I-rich and Br-rich regions under illumination⁶². Considering stability, the funneling of charges into lower-bandgap (I-rich) regions effectively reduces the maximum V_{oc} of wide-bandgap devices, resulting in a -10% decrease in V_{oc} over the first hour or so of operation^{27,63}. Illumination can also induce a similar process in mixed-cation perovskites in which cations aggregate leading to phase transitions. Density functional theory simulations indicate that cation aggregation in a FA_{0.89}Cs_{0.11}PbI₃ perovskite with a uniform distribution of cations can be triggered with relatively little energy (0.133 kJ mol⁻¹) (ref. 64).

Perovskite degradation under bias. In a solar module, partial shading by neighbouring trees, dirt or snow prevents some of the cells from generating photocurrent⁶⁵. Because all cells must pass the same amount of current for the panel to generate power, the cells that are illuminated put the shaded cells into whatever reverse bias is needed for current matching to occur. Cells in reverse bias have both short-term reversible and irreversible degradation problems.

Several interesting observations have been made on the reverse-bias behaviour of perovskite solar cells. When metal electrodes are used, the metal filaments grow all the way through the device under reverse bias to create a shunt⁶⁶. In a reverse-bias current-voltage scan, the current increases very suddenly when the shunt forms, and localized heating is observed with thermal imaging^{67,68}. The extreme heating at the shunt causes permanent damage to most layers of the cell. By contrast, this behaviour is not seen when transparent conductive oxides (TCOs) and carbon electrodes are used. In these cells, there is an exponential increase in current that starts typically between -1 V and -4 V when sweeping the voltage to increasing negative values. This current is enabled by the tunnelling of holes from the electron-transport layer due to strong band bending associated with mobile ions redistributing to screen the applied field. After current is passed through a cell in reverse bias for as little as a minute, the efficiency can be reduced by more than 50%. Much, but not all, of this efficiency recovers if the cell is exposed to 1 sun at a positive voltage for approximately 30 minutes. The rapid degradation most likely occurs because the hole density while passing the short-circuit current density in reverse bias is similar to what it would be at the MPP under 1-sun illumination, but the electron density is nearly zero. Consequently, the holes can oxidize iodide more rapidly to form smaller neutral iodine species that can move to interstitial sites, creating a vacancy in the process. With few electrons being present, the reverse reaction is slow. The iodine interstitials act as recombination centres, which greatly reduces the efficiency of the cell. During the recovery process at positive voltage, the iodine interstitials are reduced and return to the octahedral corners. The recovery only

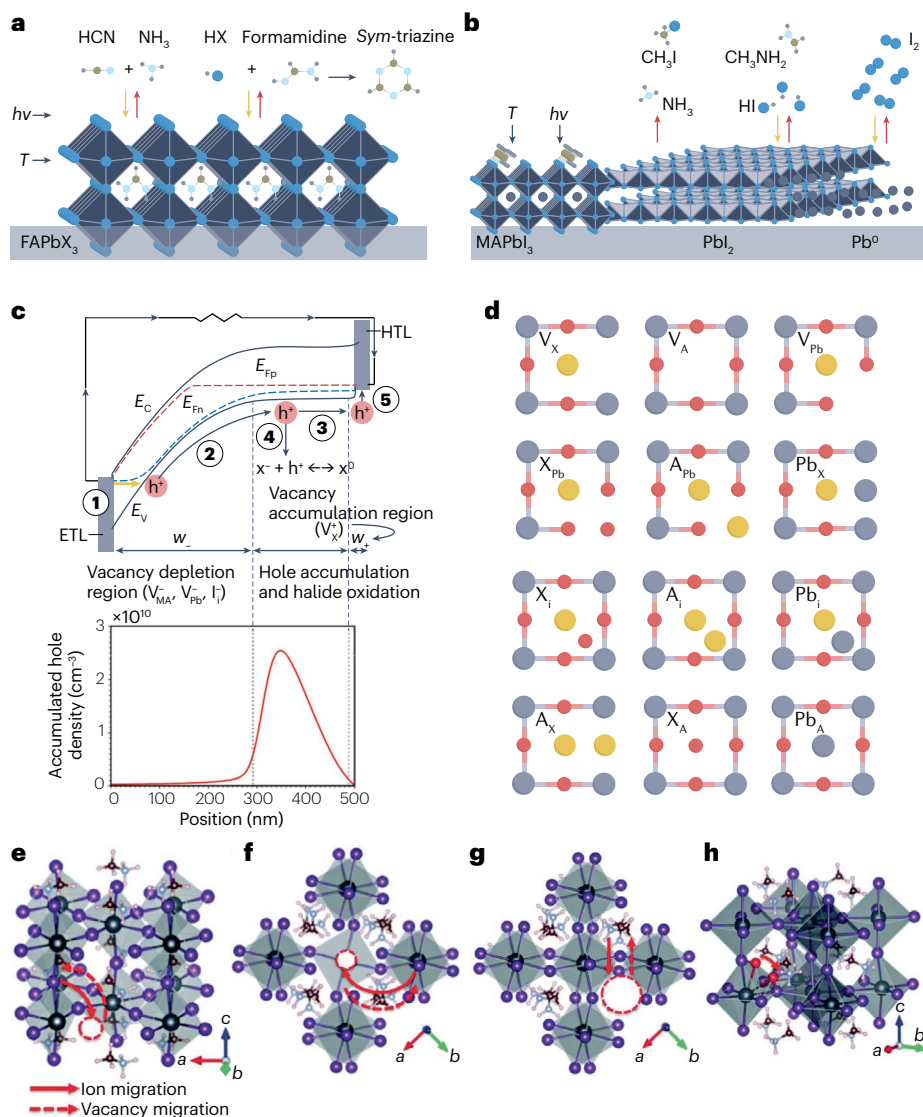


Fig. 3 | Factors affecting perovskite solar cell stability. **a**, FAPbX₃ (X = Br, I or mixed halide) degradation processes under illumination and thermal stress. Processes are indicated by different arrows (red indicates breaking processes and yellow indicates re-forming processes); irreversible processes are indicated by a single red arrow. **b**, MAPbI₃ perovskite degradation processes under illumination or thermal (same arrow convention). **c**, Energy diagram of a perovskite solar cell at -3 V showing the main processes occurring in reverse bias. (1) Hole injection: halide vacancies drift towards the hole-transport layer (HTL), resulting in favourable band energetics for hole injection via trap-assisted tunnelling at the interface between the electron-transport layer (ETL) and perovskite. (2) Hole drift: a strong electric field pulls the holes away from the space charge region. Holes accumulate at the edge of the region where there is no net electric field. (3) Hole diffusion: a high concentration gradient of holes in the bulk results in diffusion of the holes towards the HTL. (4) Halide oxidation by holes: part of the holes that build up in the perovskite bulk oxidize halides into neutral halogens. (5) Hole extraction into the HTL. In this band diagram, E_c and E_v are the perovskite conduction and valence bands, respectively. E_{fn} and E_{fp} are the hole and electron quasi-Fermi levels, respectively. W and W_s are vacancy depletion region and vacancy accumulation region, respectively. Below is the distribution of the holes injected by surface-state-assisted tunnelling and accumulating close to the edge of the vacancy-depleted region at reverse bias for a tunnelling current of 0.3 mA cm^{-2} . **d**, The 12 types of native point defects found in metal-halide perovskites. **e–h**, Diffusion paths for V_I (panel **e**), V_{MA} (panel **f**), V_{Pb} (panel **g**) and I_I (panel **h**) defects. Panel **a** reprinted with permission from ref. 52, RSC. Panel **b** reprinted from ref. 58, CC BY 3.0. Panel **c** reprinted with permission from ref. 67, Wiley. Panel **d** reprinted from ref. 75, Springer Nature Limited. Panels **e–h** reprinted with permission from ref. 76, RSC.

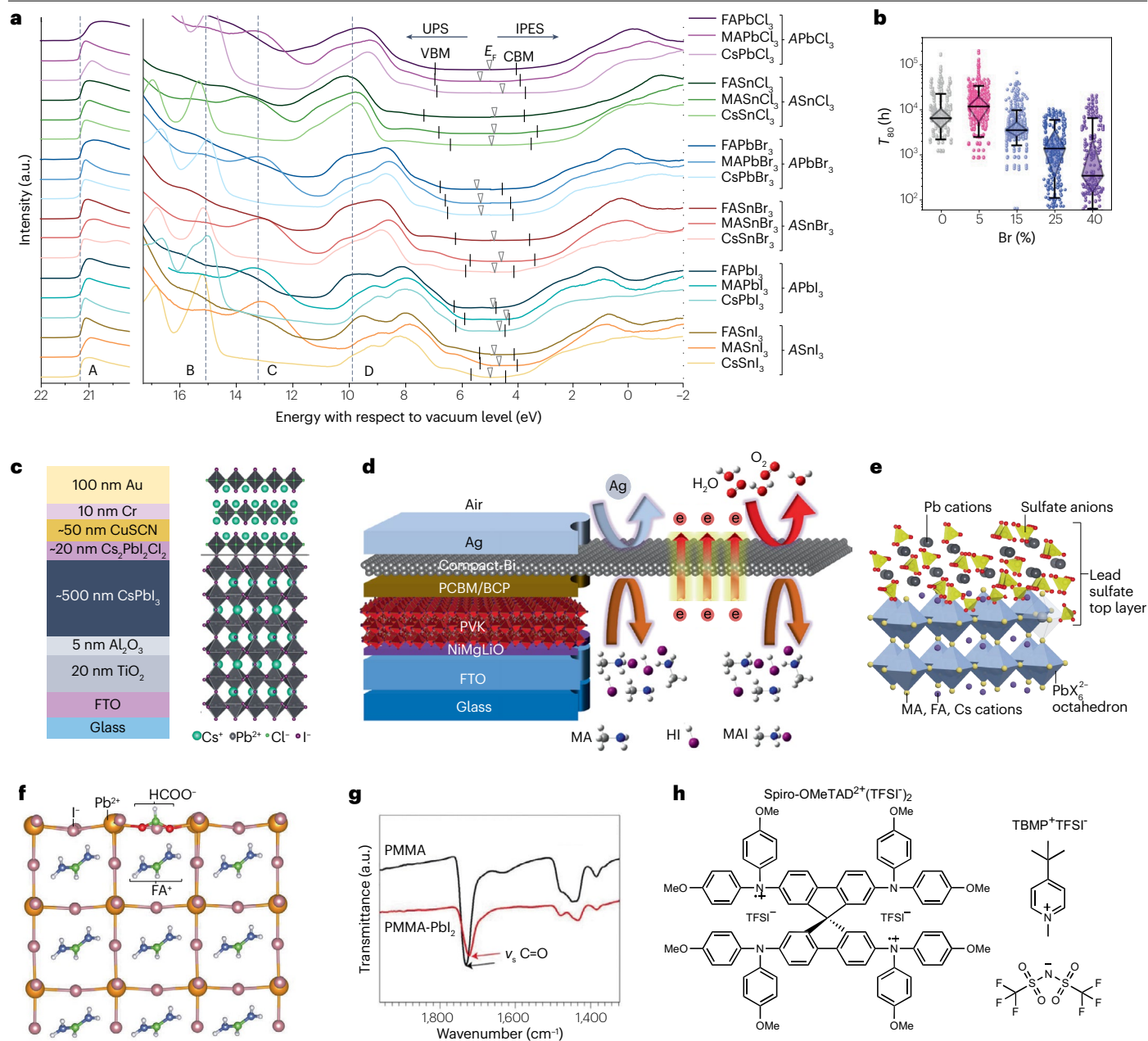
occurs, however, if the iodine species stay in the perovskite film. For this reason, internal barriers that confine the iodine to the perovskite are critically important.

If strategies for preventing degradation in reverse bias are not found, then engineering solutions for preventing degradation due to partial shading will be needed. In silicon solar panels, strings of cells are protected by a few bypass diodes that are located in the junction box. The current can be routed to the diodes through metal ribbons. The approach could be used relatively easily in panels made with perovskite–silicon tandems⁶⁹. The breakdown voltage of the tandem is the sum of the breakdown voltages of the subcells. Many tandems can be protected by one bypass diode because of the high breakdown voltage of silicon. It is not clear, however, how to use bypass diodes in thin-film panels in which cells with a length greater than a metre are separated from each other by laser scribes. If one of these cells were shaded, some of the current would have to travel as far as the length of the cell to get to the bypass diode on the edge of the panel and then

travel back. One of the electrodes would probably be a TCO, and the voltage drop over such a large distance would be excessive. It is for this reason that bypass diodes are typically not used in commercial CdTe solar panels.

It is quite possible that the degradation in reverse bias is similar in nature to what happens under 1 sun at the MPP in good solar cells that do not have weaknesses associated with poor packaging, mobile additives or reactive electrodes. Because degradation in reverse bias is approximately 100,000 times faster, it might offer an excellent way to test new materials and processing conditions rapidly to assess their impact on stability. In particular, reverse biasing can be used to generate mobile iodine in order to assess barrier quality (Fig. 3c).

Effect of defects on PSC stability. A large number of defects are generated during perovskite crystallization. Defect densities $>10^{14} \text{ cm}^{-3}$ have been reported for optimized perovskite thin films⁷⁰, most of which appear at surfaces and grain boundaries. Perovskite composition and



fabrication conditions have a substantial influence on defect type and concentration⁷¹. The primary defects affecting PSC performance are point defects at grain boundaries or surfaces⁷², including antisites (MA_{Pb} , MA_{I} , Pb_{MA} , I_{MA} and I_{Pb}), interstitials (MA_{I} , Pb_{I} and I_{I}) and vacancies (V_{MA} , V_{Pb} and V_{I})⁷³ (Fig. 3d). MHPs are defect-tolerant materials, but even shallow defects can lead to non-radiative recombination and reduce V_{OC} , albeit to a much lesser extent than traditional semiconductors. More important to stability, defects aid ion migration, meaning that defects migrate⁷⁴ (Fig. 3e–h), accumulate and propagate leading to irreversible degradation⁷⁵.

Calculations of ion migration activation energies find that halide ions contribute the most to ion migration (5 times as much as MA or Pb, for example⁷⁶). Ion migration energies vary due to measurement, perovskite composition, defect densities and PSC fabrication, but

values between 0.1 eV and 0.9 eV have been reported^{10,77–79}. Typically, these values are calculated as a function of temperature only, but ion migration is governed by temperature and illumination. For example, ion migration activation energy has been observed to decrease substantially (0.29 eV to 0.1 eV) when films are under low-intensity illumination (10 mW cm^{-2}) (ref. 80).

Defects also collect and trap carriers, inducing irreversible degradation of perovskite^{81,82}. Two deep-level trap bands, I (0.27 eV) and II (0.36 eV), have been observed in iodine-based PSCs, which correspond to iodide interstitials (I_2^- and I_3^-). Under reverse bias, a trap-filling process occurs between I_2^- and the injected holes (h^+) to form I_2^0 . I_2^0 can be oxidized to I_3^+ by valence band oxidation⁸³ or can form I_2 through a bimolecular reaction, which induces iodide loss and leads to irreversible degradation^{67,84}.

Fig. 4 | Strategies for stable perovskite solar cells. **a**, Effect of A cations on the valence band maximum (VBM) and the conduction band minimum (CBM) of different perovskite systems, obtained from ultraviolet photoelectron spectroscopy (UPS) and inverse photoemission spectroscopy (IPES) spectra. For better comparability, the curves are offset vertically, and the high-energy cut-offs are aligned at the excitation energy of 21.22 eV, marked by line A. Lines B, C and D indicate characteristic features in the density of states, corresponding to the position of Cs, MA and FA related states, respectively. The extracted positions of VBM and CBM are given by black vertical markers, and the Fermi level positions are marked by triangles. **b**, T_{80} lifetime statistics for mixed-cation, mixed-halide perovskite film with different Br concentrations. Each Br concentration contains results from 32 different cation compositions. Films were tested at 65 °C and 1-sun illumination. **c**, A perovskite solar cell (PSC) using a 2D $\text{Cs}_2\text{PbI}_2\text{Cl}_2$ layer atop the 3D perovskite active layer. **d**, PSC using a Bi interlayer and schematic of its shielding capability, prohibiting both inward and outward permeation. **e**, Schematic illustration of perovskite films protection through in situ formation of a lead sulfate top layer on the perovskite surface. **f**, Calculated crystal structure

illustrating the passivation of an I^- vacancy at the FAPbI_3 surface by a HCOO^- anion. All chemical species are shown in ball-and-stick representation. Pb^{2+} , yellow; I^- , pink; oxygen atoms, red; carbon, green; nitrogen, blue; hydrogen, white. **g**, Fourier-transform infrared (FTIR) spectra of PbI_2 -PMMA prepared by mixing polymethyl methacrylate (PMMA) with PbI_2 in a molar ratio of 1:1, and the pristine PMMA films. The arrows indicate the stretching vibration peak of $\text{C}=\text{O}$ in the two films. This shift in FTIR signal indicates the existence of a $\text{C}=\text{O}$, PbI_2 adduct that regulates crystallization. **h**, Illustration of an ion-modulated radical doping strategy for spiro-OMeTAD developed by Gao et al.¹⁶⁸. FA, formamidinium; FTO, fluorine-doped tin oxide; MA, methylammonium; PCBM, phenyl- C_{60} -butyric acid methyl ester; PVK, perovskite; TCO, transparent conductive oxide; TFSI, bis(trifluoromethanesulfonyl)imide. Panel **a** reprinted from ref. 98, CC BY 4.0. Panel **b** reprinted from ref. 22, Springer Nature Limited. Panel **c** reprinted with permission from ref. 10, AAAS. Panel **d** reprinted from ref. 128, CC BY 4.0. Panel **e** reprinted with permission from ref. 80, AAAS. Panel **f** reprinted from ref. 105, Springer Nature Limited. Panel **g** reprinted from ref. 155, Springer Nature Limited. Panel **h** adapted with permission from ref. 168, AAAS.

Effect of metal electrodes on PSC stability

Most highly efficient PSCs use a metal top electrode; gold is the most-used electrode in n-i-p devices, but copper or silver is more common for p-i-n devices. Metal electrodes are easily corroded by halides that migrate from the perovskite layer, which reduces conductivity over time. Furthermore, metal ions generated by corrosion can migrate into the perovskite, accelerating PSC degradation⁸⁵. For example, iodide on perovskite film surfaces can diffuse through CTLs to the surface of a silver electrode. The ions react with silver and form silver iodide (AgI), an insulating layer that hinders charge transport⁸⁶. The opposite is also possible. Au^+ can migrate through spiro-OMeTAD into the perovskite layer under 70 °C annealing to induce perovskite degradation⁸⁷. It is believed that under illumination, the ion migration in perovskites is reversible and does not cause permanent degradation. However, when ions, especially halides, migrate to a metal electrode and react, this causes irreversible degradation⁸⁸.

Effect of charge-transport layers on PSC stability

For highly efficient PSCs, charge-transport layers (CTLs) are indispensable. Any physical or chemical alteration of the transport layer material, the perovskite-CTL interface⁸⁹ or the CTL-electrode interface can affect the long-term stability of PSCs⁹⁰.

For example, TiO_2 , a common electron-transporting material in n-i-p PSCs, has abundant oxygen vacancy defects (Ti^{3+} sites, especially at the surface) that are effectively deep electron-donating sites. In the presence of ultraviolet light, iodide ions at the perovskite- TiO_2 interface can be oxidized into iodine, resulting in the decomposition of the perovskite⁹¹. NiOx and poly(3,4-ethylenedioxythiophene) polystyrene sulfonate (PEDOT:PSS) can undergo photo- or chemical reactions with lead halides. By contrast, pentathienoacene (PTA) and polybisphenyl(trimethylphenyl)amine (PTAA) seem inert to lead halides⁹². Currently, spiro-OMeTAD-based PSCs hold the record for single-junction PSC efficiency. Pristine spiro-OMeTAD has intrinsically low hole mobility. As such, it is necessary to increase the hole mobility by adding additives, with the most-used additives being lithium bis(trifluoromethanesulfonyl)imide (Li-TFSI) and *tert*-butylpyridine (*t*BP)^{93,94}. Although Li-TFSI and *t*BP improve PSC performance, they also bring challenges to long-term stability: *t*BP can corrode the perovskite layer through reaction with PbI_2 , and Li^+ in Li-TFSI can diffuse into the perovskite interior and decompose the perovskite structure^{95–97}.

Strategies for stable PSCs

Several stability-enhancing strategies have been developed, such as increasing ion migration energies and crystal stability through alloying, protecting the perovskite and CTLs from external stressors and passivating interfacial defects using barrier layers, decreasing defect density and suppressing ion migration using functional additives, and using robust CTLs (Figs. 4 and 5).

Perovskite phase stabilization

A-cation alloying. The A site in the perovskite crystal structure does not directly affect the band edge states as these states are formed from Pb and halide orbitals. It does, however, affect the bandgap indirectly through changes to octahedral tilting or distortion of the MX_6 framework, which changes the B-X bond length and induces VBM and CBM changes^{98,99} (Fig. 4a). Indeed, as the size and geometry of the A cation affects the bond length and bond angle of the perovskite crystal, it has great influence on the crystal stability and the electrical properties of the perovskite.

FAPbI_3 has a narrower bandgap (1.48 eV) than that of MAPbI_3 (1.6 eV) (Fig. 4a), so FAPbI_3 -based perovskites are preferred as their bandgap is closer to optimal for maximum device efficiency (1.34 eV)¹⁰⁰. Indeed, the majority of high-efficient PSCs are FAPbI_3 -rich recipes. However, as the tolerance factor for FAPbI_3 is close to 1, FAPbI_3 will naturally lose its cubic phase at room temperature. To stabilize α - FAPbI_3 , partial replacement of FA^+ with smaller-radii cations (MA^+ , Cs^+ , Rb^+) is used. The introduction of MA^+ enables the perovskite to obtain its α -phase at a lower temperature than pristine FAPbI_3 (ref. 101), but is generally thought to decrease the light and thermal stability of FAPbI_3 perovskite as MA^+ is more volatile than FA^+ (ref. 102). Counterintuitively, up to -10% MA incorporation has been shown to improve stability^{22,103}.

Partially replacing FA^+ with smaller-radii cations such as Cs^+ not only stabilizes α - FAPbI_3 but also improves thermal stability due to the strong chemical bonding between Cs^+ and $[\text{PbI}_6]^{4-}$. $\text{Cs}_{0.1}\text{FA}_{0.9}\text{PbI}_3$ has shown better photo- and moisture stability than FAPbI_3 . This stability is attributed to the introduction of Cs^+ , which shrinks the perovskite crystal cubo-octahedral volume and increases the A-site-I bond energy¹⁰². To further improve stability, complex mixed-cation systems have also been explored, such as CsMAFA and RbCsMAFA . Adding more elements increases the entropy of mixing to stabilize the perovskite phase^{40,104}.

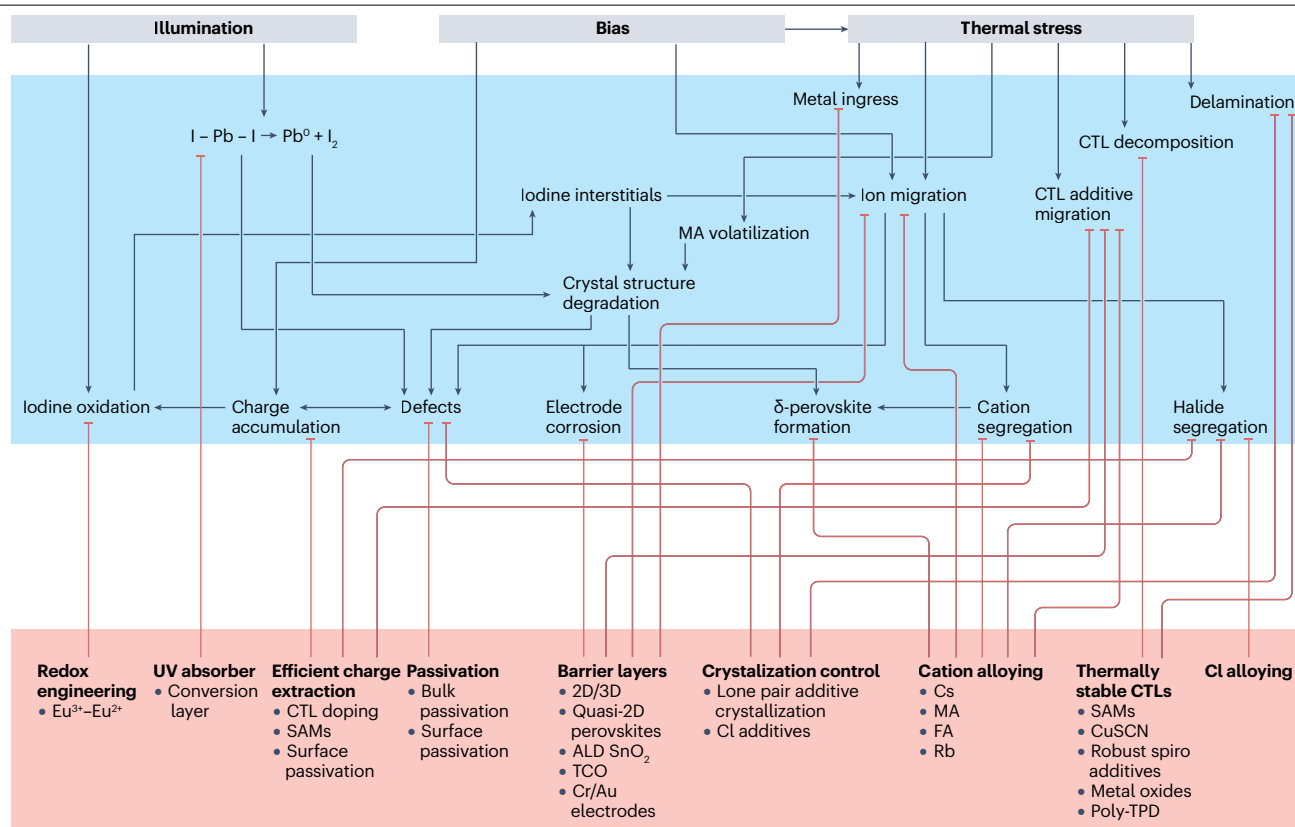


Fig. 5 | Overview of perovskite solar cell stability. The grey boxes represent causes and the blue box effects. The red box represent mitigation strategies and specific implementations. 2D/3D, 2D/3D heterostructure solar cell; ALD, atomic layer deposition; CTL, charge-transport layer; CuSCN, copper thiocyanate;

FA, formamidinium; MA, methylammonium; poly-TPD, poly[*N,N'*-bis(4-butylphenyl)-*N,N'*-bis(phenyl)benzidine]; SAMs, self-assembled monolayers; spiro, 2,2',7,7'-Tetrakis(*N,N*-di-*p*-methoxyphenylamine)-9,9'-spirobifluorene; TCO, transparent conductive oxide.

Various other methods to stabilize α -FAPbI₃ have been discussed in the literature (for example using pseudo-halides¹⁰⁵ or using methylenediammonium³⁸ or isopropylammonium¹⁰⁶ as additives). For many of these, it is unclear whether perovskite alloying has occurred. Including a small amount of Cs⁺ or MA⁺ to stabilize FAPbI₃ results in a non-cubic perovskite crystal structure with an octahedral tilt of $\sim 2^\circ$ at room temperature¹⁰⁷. This α -phase stabilization is not exclusive to alloying additives, as it can also be achieved by additives at the perovskite grain boundaries. Moreover, if alloying cations are unevenly distributed, nanoscale local regions of perovskite can lose their 2° tilt, forming regions of hexagonal polytypes that expand over time, and resulting in δ -phase formation¹⁰⁷. Increasing the annealing temperature or the annealing time results in more homogenous mixing of A-site cations and higher PSC stability¹⁰⁸.

Introducing large-radius cations such as PEA⁺ or BA⁺ into FAPbI₃ perovskite is another important strategy to improve perovskite stability¹⁰⁹. The exchange of large-radii cations leads to the collapse of the 3D perovskite crystal structure and the introduction of a quantum-confined, usually 2D structure¹¹⁰. If large A cations are mixed directly into the perovskite precursors, then a mixed, quasi-2D film containing various thicknesses of 2D and quasi-2D perovskite tends to form. This mixed film has enhanced stability due to an increased hydrophobicity and the presence of large-cation layers hindering ion

transport, but lower efficiency due to an increased bandgap, decreased mobility and higher exciton binding energies¹¹¹. Applying 2D perovskite atop the bulk 3D to passivate the surface and block ion transport out of the cell can be used to remediate these issues.

A summary of operational stability enhancements from A-site alloying is found in Supplementary Table 3.

X-anion alloying. Unlike the A-site cation, the X-site anion directly contributes to the band edge of the perovskite. Replacing I⁻ with Br⁻ or Cl⁻ can tune the bandgap, regulate perovskite grain growth and improve the chemical stability of perovskite films. Partially replacing I⁻ with Br⁻ can suppress the δ -phase formation of FA-rich or Cs-rich perovskites^{112,113}, and perovskites with 5% Br have shown better T_{80} stability²² (Fig. 4b).

The introduction of Cl⁻ into perovskite precursors can substantially improve the quality, morphology¹¹⁴ and crystallinity of majority iodine perovskites, although whether Cl alloys into the perovskite is still debated and probably depends on various factors¹¹⁵. Introducing MAcl into FAPbI₃ perovskite is one of most popular methods to prepare stable α -FAPbI₃ films¹¹⁶, as the Cl-containing perovskite system is more thermodynamically stable and the introduction of MA shrinks the volume of the FAPbI₃ crystal, which also improves the stability of the perovskite structure¹¹⁷. In wider-bandgap perovskites that contain

important amounts of Cs and Br, the lattice parameters are smaller, and more chlorine can be incorporated on the X site¹¹⁸. These compounds are more stable against light-induced phase separation.

All-inorganic perovskites. All-inorganic perovskites (Cs,Rb)Pb(I,Br)₃ are more thermally stable than their hybrid counterparts. Because the bandgaps of all-inorganic perovskites are wider than those of FA-based perovskites, they are typically considered for tandem applications. The narrowest bandgap available is for CsPbI₃ (~1.7 eV), which is well matched for Si/perovskite tandems. Like for FAPbI₃, the photoactive phase of CsPbI₃ is difficult to maintain at room temperature. Many strategies have been developed to stabilize α -CsPbI₃: A-cation alloying, X-anion alloying and additive engineering. For example, 18-crown-6 ether has been used as an additive to stabilize the cubic phase of CsPbI₃, and devices maintained ~91% of their initial PCE after 1,000 h in N₂ at 85 °C (ref. 119). Polyvinylpyrrolidone (PVP) has likewise been used, with PSCs showing no visible decrease in PCE after 3,500 h at 35 °C, and decreasing by only 20% after 2,100 h under continuous illumination at 110 °C (ref. 10) (Fig. 4c).

Although highly thermally stable, all-inorganic PSCs tend to suffer from lower efficiencies than their hybrid counterparts. These lower efficiencies have been attributed to high defect densities induced by fast crystallization, resulting in short carrier lifetimes and low photoluminescent quantum efficiency¹²⁰. To improve efficiency, dimethylammonium has been used to stabilize the β -phase CsPbI₃ and to control perovskite crystallization¹²¹, and surface treatments have been used to reduce surface defect densities¹²⁰. Using a CsF surface treatment, an all-inorganic PSC with a 1.74-eV bandgap achieved 21% PCE¹²², ~75% of the thermodynamic limit. The highest-performing hybrid perovskites can achieve >80% of this limit¹²³, so there is some room for improvement. The carrier lifetimes recorded for these all-inorganic films are roughly one-quarter that of the highest-performing hybrid films (~700 ns compared with ~2,900 ns), suggesting that defect densities can be further reduced.

Barrier layers for stable PSCs

Barrier layers are thin layers deposited at interfaces within a PSC to improve efficiency and stability. These layers can protect from external stressors (such as moisture or ultraviolet light), passivate interface defects, block ion migration and inhibit interfacial chemical reactions¹²⁴.

CTL–electrode interface barrier layers. The main function of a barrier layer at the CTL–anode interface is to block the migration of perovskite and metal ions crossing the interface without reducing conductivity. There are various barrier layers in use (metal oxides¹²⁵, graphene¹²⁶ and organic self-assembled monolayers (SAMs)¹²⁷). An illustrative example is Au or Ag infiltration in PSCs aged at high temperature. Introducing 10 nm of chromium between a Au anode and the HTL entirely mitigates Au infiltration into the perovskite layer after 75 °C ageing under light, suggesting that Cr is an effective barrier to metal ion migration⁸⁷. Alternatively, bismuth interlayers can protect metal electrodes from iodine corrosion or perovskite ingress, preventing all detectable Ag infiltration into the perovskite layer after ageing at 85 °C for 100 h under light. Bismuth-protected devices maintain 95% and 97% of their initial PCE after 500 h under 85 °C and light soaking conditions, respectively¹²⁸. TCO electrodes such as indium tin oxide or indium zinc oxide can also reduce ion infiltration and increase stability. The most effective TCO layers are fabricated by sputter deposition, but

high-energy sputtering can easily damage other layers. Atomic layer deposition (ALD) can remediate this issue as has been demonstrated with SnO₂ in both perovskite tandems and single-junction devices¹²⁹. SnO₂ can act as a barrier layer in its own right, or in conjunction with a sputtered TCO layer¹³⁰.

Alternatively, one can replace metal electrodes with carbon electrodes to avoid degradation due to electrode corrosion or metal ion migration. Interface barrier layers are often used to improve the contact or energy-level alignment between the carbon electrode and the perovskite layer, partially offsetting the efficiency decrease caused by the carbon electrode¹³¹.

Perovskite–CTL interface barrier layers. The high surface defect density of perovskite films accelerates degradation while corrupting other layers through ion migration. Barrier layers that can passivate surface defects and block ion migration are highly attractive. Various materials have been demonstrated as perovskite–HTL interface barrier layers, including 2D materials¹³², quantum dots¹³³, metal oxides^{134,135}, functional polymers¹³⁶, small molecular compounds^{137,138} and ammonium salts^{139,140}. The most popular passivation/barrier layer in modern PSCs is a layer of 2D perovskite, with several reports of good operational stability at ≥ 60 °C in 2022 alone^{9,10,141,142}. For example, pure-phase 2D perovskite single crystals dispersed in acetonitrile were coated onto 3D perovskite to form a highly controllable 2D/3D interface. An advantage of 2D perovskite is its relatively high conductivity compared with other insulating barrier layers, meaning that layers up to 80 nm in thickness have been used without resistive losses¹⁴³. Using a conformal 50-nm-thick 2D layer, devices with >23% PCE were fabricated, which also demonstrated $T_{99} > 2,000$ h following the ISOS-L1 (MPP, 60 °C) protocols, a large increase compared with control devices (T_{80} of ~500 h)⁹ (Fig. 4d). Depositing CsCl onto CsPbI₃ has been used to form an all-inorganic 2D/3D heterostructure (Fig. 4c). This strategy increased the barrier to ion migration in PSCs by a factor of 2 and increased the T_{80} stability (ISOS-L3) by a factor of ~3 at 110 °C to >2,000 h. Device stability was measured at several temperatures and used to extrapolate the T_{80} lifetime at 35 °C to be ~5 years¹⁰.

Using non-2D perovskite barrier layers, such as inorganic lead sulfate, also leads to impressive stability results. For example, although the ion migration energy of control devices is ~0.1 eV, no transition to ionic transport was observed in lead-sulfate-capped PSCs below 55 °C (Fig. 4e). Using this technique produced >20% PCE cells with a $T_{97} > 1,000$ h at 65 °C (ISOS-L3)⁸⁰. In addition, hybrid material ferrocenyl-bis-thiophene-2-carboxylate (FcTc₂) strongly binds to the perovskite surface and effectively prevents surface MA⁺ from escaping the film under light (1 sun) and heat (85 °C). Devices with PCE ~24% exhibited a T_{98} of 1,500 h (ISOS-L1, room temperature)¹⁴⁴.

Additive engineering

Perovskite grain boundaries and surfaces are vital to the stability of PSCs as they contain the majority of defects, and these defects tend to propagate over time^{15,145}. Reducing the density of defects at grain boundaries inhibits degradation and ion migration. Both functional materials and solvent additives can regulate crystallization to increase grain size and passivate grain boundaries¹⁴⁶. This section investigates the effects of salts and organic compound additives on PSC stability.

Additive salts. A popular set of additives are organic ammonium halide salts. These additives exist in different formations at perovskite grain boundaries and surfaces. Some ammonium species form

low-dimensional perovskites, which improve PSC stability by inhibiting ion migration^{147,148}. Alternatively, organic cations with large molecular volumes, such as $\text{NH}_3\text{I}(\text{CH}_2)_8\text{NH}_3\text{I}$, cannot form low-dimensional perovskite but can heal surface defects through hydrogen bonding, X-site vacancy passivation (halogens) and NH_3 groups binding to A-site vacancies or Pb–X antisite defects^{2,149}.

Ionic liquids (ILs) are increasingly used to improve the stability of PSCs through three mechanisms¹⁵⁰. First, the ionic liquid reacts with PbI_2 to form an intermediate complex, which increases the activation barrier to perovskite nucleation growth, thus increasing grain size and reducing grain-boundary density. Next, the formation of hydrogen bonds between the ionic liquid and MA^+ or FA^+ inhibits relevant ion migration, thereby increasing PSC stability. Finally, ionic liquids have various functional groups that can passivate different perovskite surface defects, suppress ion migration and retard lead oxidation and the compositional segregation of perovskites. Formamidinium formate (FAHCOO) was used to fabricate >25% PCE PSCs with a T_{80} of 400 h (compared with 150 h for control devices)¹⁰⁵ (Fig. 4f).

Organic compounds as additives. Organic compounds are some of the most commonly used perovskite additives. These materials contain N-donor, S-donor or O-donor groups. The electron lone pair of these groups can react with lead halide precursors to form complexes that, similarly to ionic liquids, retard the nucleation rate of perovskites and reduce the density of grain boundaries¹⁵¹. Concurrently, these electron donors effectively passivate perovskite defects, thereby improving the stability of PSCs. Some oligomer additives can also be used as linkers between grains to improve grain-boundary stability¹⁵². Besides the above functions, conjugated organic additives can modify energy-level alignment to promote carrier extraction at interfaces, inhibiting the accumulation of charges and improving stability^{153,154}.

A notable example is PMMA, which contains C=O functional groups. When PMMA is added to a perovskite precursor solution, an intermediate adduct is formed between C=O and PbI_2 (Fig. 4g). The intermediate adduct formation effectively regulates the crystallization process of the perovskite and increases the perovskite crystal size considerably¹⁵⁵. As another example, the theophylline molecule contains N–H and C = O functional groups. The N–H groups can form a hydrogen bond with iodine and assist the interaction of C=O with antisite Pb (lead) defects on the perovskite surface leading to suppressed ion migration, devices with theophylline showing better stability under operation¹⁵⁶.

It should be noted that there is currently no standard for evaluating which type of additive is better, nor is there any authoritative comparative study indicating additive hierarchy. At present, there is consensus in the field of PSCs that materials containing S, N, O or P atoms can form adducts through Lewis acid–base reactions with lead atoms, which regulate perovskite crystallization and reduce defects in perovskite films^{151,157}. However, how best to compare the efficacy of different additives remains a problem for the field.

Robust transport materials and relevant additives for stable PSCs

For p–i–n PSCs and p–i–n-based tandem devices, the most commonly used HTLs (PTAA, NiOx and SAMs) have shown good stability^{37,158}, and ALD SnOx as ETL can show good stability and high efficiency. For n–i–p devices, SnO₂ seems to effectively solve the instability induced by TiO₂ (ref. 159), but resolving the instability induced by spiro-OMeTAD while ensuring higher efficiency is a challenge^{160–162}. Here we highlight work

on improving the stability of spiro-OMeTAD and on the development of additive-free HTMs, as many HTM additives cause degradation when they diffuse into the perovskite.

PSCs with record efficiency have consistently been fabricated in the n–i–p structure. Spiro-OMeTAD is the best-performing HTM for these cells in terms of efficiency, but it requires dopant additives to increase its mobility, and stability problems caused by additives in spiro-OMeTAD have resulted in consistently poor stability from record devices (Fig. 2c). To solve or offset the negative effects of spiro-OMeTAD additives, several strategies have been used, including reducing the use of additives¹⁶³, introducing Li⁺ complex agents to inhibit Li⁺ migration^{164,165}, introducing substances with hydrophobic groups¹⁶⁶ and developing new doping regulators¹⁶⁷. For example, stable organic radicals have been used as dopants along with ionic salts as doping regulators to oxidize spiro-OMeTAD and increase its hole mobility (Fig. 4h). The efficiency of devices made using this method exceeded 25% PCE, with the T_{80} of unencapsulated devices aged at 70 °C (measured in an N₂ glovebox) increasing from ~264 h to ~796 h (ref. 168).

Many researchers are focused on the development of additive-free HTMs, including organic small-molecule materials^{93,169}, polymers¹⁷⁰ and inorganic hole-transport materials¹⁷¹. For example, PSCs using compact, conformal CuSCN as a HTL have recorded PCEs exceeding 20% and show excellent stability at 60 °C over 1,000 h (ref. 172). Furthermore, the efficiency of additive-free, polymer-HTL-based PSCs has exceeded 24% and maintain 80% original efficiency after ~1,000 h under 1-sun irradiation (white LED, 100 mW cm^{–2}, 25 °C, under N₂)¹⁷³.

Accelerated lifetime testing

It is impractical to use decades-long protocols for stability testing. Hence, correlation between accelerated testing protocols, degradation models and real-world outdoor testing were used to establish the IEC standards used to certify Si PV panels¹⁷⁴. These standards do not extend to perovskite PV as the acceleration factors used to correlate simulated and real-world testing are different³⁵. Additionally, perovskite PV demonstrates behaviour unknown in Si PV related to ion migration and electrochemistry, such as performance recovery under dark conditions^{175,176}.

There are two purposes to stability testing. The first is to identify the rough operational stability of a device in comparison to a control or another study – this type of reporting is common across PSC research. The second is to identify the failure modes of devices and to develop a model for how devices degrade with time. Considering that many PSCs have been reported with little to no degradation under ISOS-L1 conditions for >1,000 h (refs. 9,141,177), accelerated testing and modelling of degradation modes is becoming ever more important. This concept has been used successfully in organic photovoltaics (OPV) and in Si PV where degradation is assumed to derive from a set of individual activation energies that interact multiplicatively¹⁷⁸, but has seldom been used in perovskite PV.

In OPV and Si PV, thermally induced decay is modelled using the Arrhenius equation, to which light intensity is simply assumed to apply multiplicatively. For example, for an arbitrary decay function with some number of constant decay rates k ,

$$\text{PCE}(t) = f(k_1, k_2, k_3, \dots, t) + C$$

where each decay rate represents a different decay mode in the material (such as halide migration, cation diffusion or B-site oxidation). Traditional models use

$$AF = \frac{k_{acc}}{k_{ref}} = e^{\frac{E_a}{k_B} \left(\frac{1}{T_{ref}} - \frac{1}{T_{acc}} \right)}$$

as the acceleration factor for thermal degradation, where E_a is the activation energy for thermal degradation, and T_{ref} and T_{acc} are reference and elevated temperatures, respectively¹⁷⁸. To include light intensity, a factor is multiplied such that

$$AF_{L+T} = AF_L AF_T = \frac{I_{acc}}{I_{ref}} e^{\frac{E_a}{k_B} \left(\frac{1}{T_{ref}} - \frac{1}{T_{acc}} \right)}$$

where I_{acc} and I_{ref} represent the light intensity of accelerated and reference testing, respectively. Depending on the decay mode, one might find that only some decay rates are affected by heat and others by illumination intensity¹⁰. For example, a simple model for N_2 -encapsulated $CsPbI_3$ PSCs that considers only one light intensity has been reported. The PCE of devices with and without a 2D capping layer degraded according to a biexponential relationship, with two decay rates. Interestingly, both decay rates followed the same relationship with temperature, so were both derived from the same activation energy. Fitting data to this simple model found good agreement, and the lifetime of cells operated at 35 °C could be predicted to be >5 years (Fig. 6a,b). This result is even more impressive when considering that these cells were under constant 1-sun illumination, which is roughly 5 times the average irradiation experienced by a typical real-world panel, suggesting a similar lifetime to that of established PV technology²⁵.

An interesting finding from this work is that the common fast initial decay, often classified as ‘burn-in’, could be generated by the same mechanism that causes slow long-term decay. What is still unclear, however, is why ion migration has two different decay modes and to what extent each of them is reversible.

A more comprehensive model for $MAPbI_3$ that includes humidity, oxygen, light intensity as well as temperature has been developed¹⁷⁹. In this model, the light intensity acceleration factor is $(I_{acc}/I_{ref})^{0.7}$, so less important than the linear multiplicative factor used for established PV, but still indicates that high light intensities could substantially speed up degradation testing. Little work has been done on accelerated testing using high light intensities, although two reports demonstrate stability up to 10-sun intensity (T_{90} ~ 150 h, Fig. 6c) suggesting this could be a useful parameter to accelerate testing timescales^{180–182}.

Another key consideration when selecting accelerated testing scenarios is the effect of cycling ageing conditions as, in real-world use, a large discrepancy between light-cycled devices (12 h light, 12 h dark) and continuously illuminated devices will occur. For example, in one study, a light-cycled device yielded an average efficiency of 96% initial PCE across 250 h, compared with a continuously illuminated cell that averaged only 88% initial efficiency (Fig. 6d). In the same study, temperature-cycled devices (65 °C to –10 °C every 2 h) degraded at a rate in between devices held at either –10 °C or 65 °C (ref. 175). Monitoring the performance of a device subjected to simulated weather conditions from two days of each month in the year resulted in several interesting observations. Over the course of each day, the PSC degraded substantially relative to what was expected from single MPP measurements at each light intensity and temperature. Initially, the degradation was reversible and the device completely recovered by the second day, but the proportion of reversible decay reduced over time, and after the first week most degradation was irreversible. Exposure to high light intensities seemed to dominate PSC degradation rather than high temperatures. Understanding what causes reversible

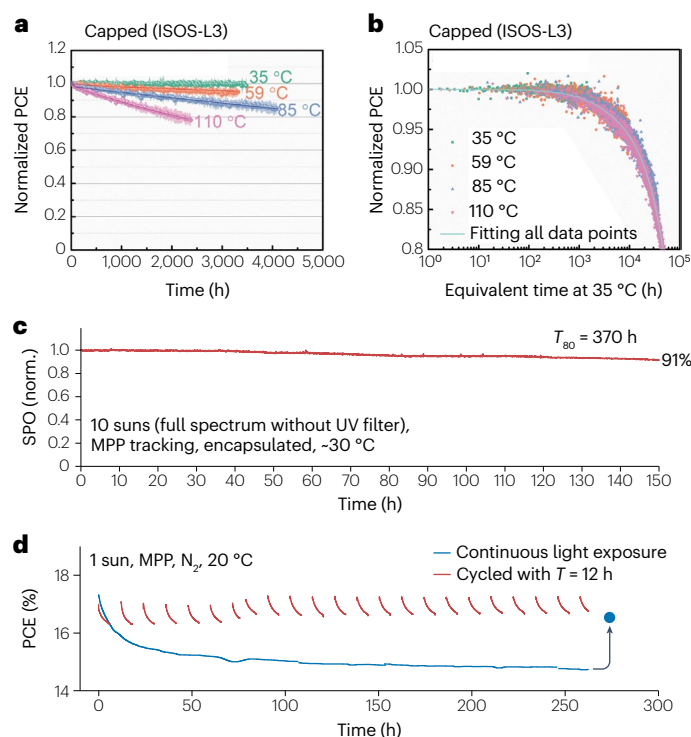


Fig. 6 | Accelerated ageing. **a**, Results from accelerated ageing at different temperatures for $CsPbI_3$ all-inorganic perovskite solar cells (PSCs). **b**, Modelling these data using an Arrhenius factor fitting to calculate an acceleration factor, suggesting that the cell aged at 35 °C should have a T_{80} > 5 years. **c**, Maximum-power-point (MPP) tracked PSC data at 10-sun intensity using a fluorine-doped tin oxide/SnO₂/phenyl-C₆₁-butyric acid methyl ester/FA_{0.83}CS_{0.17}PbI_{2.7}Br_{0.3}/polybisphenyl(trimethylphenyl)amine/Au structure. **d**, Data comparing MPP-tracked PSCs using constant 1-sun intensity and light cycling. The light-cycled device produced 96% of its initial efficiency on average throughout the 250-h test, whereas the device under constant illumination produced only 88% of its initial efficiency on average. PCE, power conversion efficiency; SPO, stabilized power output. Panels **a** and **b** reprinted with permission from ref. 10, AAAS. Panel **c** reprinted from ref. 180, Springer Nature Limited. Panel **d** reprinted from ref. 175, Springer Nature Limited.

and irreversible change to PCE is a pertinent question for future testing. Even this small pool of results suggests that many different testing conditions need to be applied, modelled and compared to build up a reliable testing protocol.

Testing with multiple stressors at once is essential to obtain reliable results. For example, in one study, encapsulated $Cs_{0.05}FA_{0.8}MA_{0.15}Pb(I_{0.85}Br_{0.15})_3$ cells showed no degradation after 2,000 h of IEC standard damp-heat testing. When subjected to MPP tracking under illumination, however, the same cells degraded to 80% of their initial PCE in a few tens of hours under illumination at an elevated temperature (45 °C)¹⁸³. To combat this discrepancy, researchers should compare different ageing conditions individually and in combination. First, researchers should find an encapsulation technique that can allow cells to pass the IEC61215:2016 damp-heat test (85 °C/85% relative humidity), and then operational testing under different light and temperature conditions should be examined. These conditions should include varying light

intensity (0.1 to 10 suns, with and without ultraviolet light), temperature cycling (-40°C to 85°C) and light cycling (0 to 1 sun).

Perspective and conclusion

For many years now, stability rather than efficiency has been the most pressing issue in PSC research. In this time, great progress has been made, and the field has produced a toolbox with which we can fabricate highly stable PSCs. Indeed, it should come as no surprise that the strongest operational stability yet reported (ref. 10) used PSCs that are all-inorganic (high-temperature stability), have Cr/Au electrodes, and use 2D perovskite and Al_2O_3 barrier layers. Clearly the field is advancing, and by using many stability-enhancing techniques in parallel, multiyear operational stability has been achieved.

A few final pieces are needed to demonstrate long-term stability: accurate models that allow faster accelerated ageing on a wider range of PSCs, methods to increase predicted T_{80} stability to >25 years and techniques to upscale stable methods for large-area modules. Here, we give our view on how best to approach these challenges.

Modelling degradation

In theory, one can accurately model the degradation from multiple stressors and degradation modes by patiently examining stability curves and fitting them to increasingly complex functions. Instead, it is easier to work on a PSC architecture that is already reasonably robust, which reduces the number of potential degradation modes and simplifies the model. Encapsulating cells in multiple layers and/or using glass–glass lamination seems to negate environmental effects. Then, by using stable transport layers and conformal blocking layers, we can focus on the perovskite film. Limiting temperatures to $<90^{\circ}\text{C}$ for FA-based perovskites and to $<300^{\circ}\text{C}$ for inorganic perovskites can restrict unrealistic thermal decomposition. Previous testing suggests that only a few degradation modes are present after such isolation: ion drift, decomposition of PbI_2 to Pb^0 and I_2 gas, cation segregation and halide segregation. Although still a complex system, these processes are governed by ion migration energies and light-induced decomposition, so only a small number of measurements are required. We expect sophisticated models that capture this behaviour to appear soon. Once established, they can be used to accelerate testing protocols further, using high illumination intensities and high temperatures.

Methods to increase perovskite stability

Although impressive increases to PSC stability have been achieved, these results do not yet meet the requirements for commercial application. To further increase stability, additional techniques should be considered in combination with other strategies. We suggest some underappreciated methods below.

Single-crystal PSCs. Single-crystal perovskites are archetypal materials for investigation as they have no grain boundaries and have defect densities several orders of magnitude lower than their polycrystalline counterparts^{184–186}. As such, single crystals demonstrate suppressed ion migration, charge accumulation and defect-induced phase transitions¹⁸⁷. The relatively small number of reports on single-crystal PSCs have shown that single crystals are superior to polycrystalline films at withstanding humidity and heat¹⁸⁸. Through modified HTL–perovskite interface, MAPbI_3 single-crystal devices with 22.1% PCE have been achieved. These devices maintained 90% of their initial PCE after 1,000 h in air, compared with controls maintaining only 61%¹⁸⁹. Given their inherent advantages, we expect single-crystal PSCs to display

impressive stability under operation, but more work is necessary to conclude on the matter.

Quasi-single-crystalline PSCs. We currently lack methods to upscale single-crystal PSCs, and cell areas are limited to $\sim 0.1\text{ cm}^2$. Quasi-single-crystal film growth (that is, the growth of films composed of micrometre-scale single crystallites through seeded growth) represents a promising middle ground between traditional thin-film processing and single-crystal PSCs. For example, perovskite thin films have been fabricated by first forming a repeating array of 5- μm crystals spaced 5 μm apart, seeding the growth of a continuous film with grain size exceeding 3 μm (ref. 190). Stability tracking of encapsulated devices (FTO/ SnO_2 /perovskite/PEAI/spiro-OMeTAD/Au, where FTO is fluorine-doped tin oxide) at room temperature under AM1.5 1-sun intensity (ISOS-L1) demonstrated a T_{80} lifetime of 200 h for control devices and a T_{90} of 2,000 h for the seeded growth films (initial PCE $\sim 24\%$). Optimized PSC architecture could further improve the stability of quasi-single-crystal devices.

Down-conversion layer. Because shorter-wavelength light is the most detrimental to perovskite stability, introducing a barrier layer to absorb ultraviolet light is an obvious way to enhance PSC stability. However, this approach also reduces the short-circuit current density J_{sc} (ref. 56). Using a down-conversion material as a barrier layer solves this issue by absorbing ultraviolet light and converting it into longer (visible) wavelengths. Although this concept has attracted some attention⁶², it has mostly been applied to non-optimized MAPbI_3 cells in an attempt to increase the current density, rather than to achieve higher stability. Conversion layers with photoluminescent quantum efficiency $>80\%$ induce only about a 2% increase in J_{sc} , and this small improvement in current probably does not justify the added cost of the additional layers. These down-conversion layers, however, could be integrated in cells to improve their stability. For example, MAPbI_3 cells using a CsPbCl_3 :Mn quantum dot conversion layer yielded a fourfold increase in stability under ultraviolet light¹⁹¹. The use of this method in a highly stable cell architecture is an exciting prospect.

Upscaling stability

Most of the methods described herein were demonstrated on cells with $<1\text{ cm}^2$ active area. Upscaling this to modules involves a change in approach, moving away from spin coating to techniques such as blade coating, slot-die coating, screen printing or vacuum evaporation. These processes have a whole host of new engineering challenges to produce PCEs and stabilities comparable with small-scale, spin-coated devices. For solution-processed films, volatile solvents such as DMSO are usually used, but unevaporated trapped DMSO can escape to produce voids between the perovskite and CTLs. This issue is accentuated in large-area devices causing rapid degradation under light and heat. Using non-volatile carbonylhydrazide to replace DMSO can overcome this issue. Carbonylhydrazide is reported to remain in films even after long-term exposure to 1-sun illumination at 60°C , and 50- cm^2 modules using carbonylhydrazine have been fabricated¹⁹² with efficiencies approaching 20% PCE and T_{85} lifetimes of 1,000 h under 1-sun illumination at 50°C . Even more encouraging is the report of a $T_{97} > 1$ year for 20- cm^2 devices under light cycling, produced by replacing DMSO with another less volatile agent (diphenyl sulfoxide)¹⁹³.

In monolithically integrated panels, P2 laser scribing can leave thermally damaged perovskite near the edges of the scribe. Furthermore, if metal is used to fill the via (the trench created in the scribing

process), the direct contact with the perovskite can cause rapid degradation because halides react with metal¹⁹⁴. Making singulated cells is advantageous to avoid the use of laser scribes. This approach keeps barriers intact, avoids passing metal through the perovskite, avoids thermal damage associated with laser scribing, and allows the binning of cells by efficiency. If singulated cells are connected with a wire, current can be routed around a cell and through a bypass diode to prevent the cell from being pushed into reverse bias if it is in the shade while the other cells are illuminated. Perovskite–silicon tandems naturally take this approach.

These promising results suggest that highly stable modules are achievable. We expect a continued focus on module stability as the field progresses towards commercial stability.

Conclusion

Despite being a persistent problem in perovskite PV, stability has improved by orders of magnitude in the first decade of mainstream perovskite PV research. With the introduction of various stability-enhancing methods, the operational stability of PSCs is maturing beyond practically achievable testing lifetimes. The introduction of degradation modelling and lifetime estimation has suggested that some current PSC configurations could already have lifetimes similar to established PV. Currently, PSC researchers are using the ISOS protocols adapted from OPV research. Further degradation analysis and comparison with outdoor testing is critical to generate perovskite-specific protocols for calculating cell lifetimes and to understand how to improve stability further. Because many factors cause PSC degradation, solving stability issues using only one method is unrealistic, and the synergistic use of various stability-enhancing methods is the way forward. Considering what has already been achieved and the continued huge effort expended in this field, we expect that perovskite solar modules that retain 90% of their initial performance after 25 years of operation will soon emerge.

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Author contributions

H.Z. and S.T. contributed equally to the article. H.Z., S.T., M.N.L., S.M. and B.C. researched data for the article. All authors contributed substantially to discussion of the content. All authors

wrote the article. O.M.B., H.Z., S.T., M.D.M. and E.H.S. reviewed and/or edited the manuscript before submission.

Competing interests

M.D.M. is an adviser to Swift Solar. The other authors declare no competing interests.

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